

Statistics Unit 5: Multiple Regression Lecture Notes

Simple regression:

$$y = b_0 + b_1x_1 + e$$

Multiple regression:

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + e$$

y = Client wellbeing

x_1 = Alliance

x_2 = Therapist positive regard

x_3 = Client agreeableness

Predicted y : $\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3$

Error: $y - \hat{y}$

Calculate values for b_0, b_1, b_2, b_3 (etc.) that minimize the sum of squared errors. $\Sigma(y - \hat{y})^2$

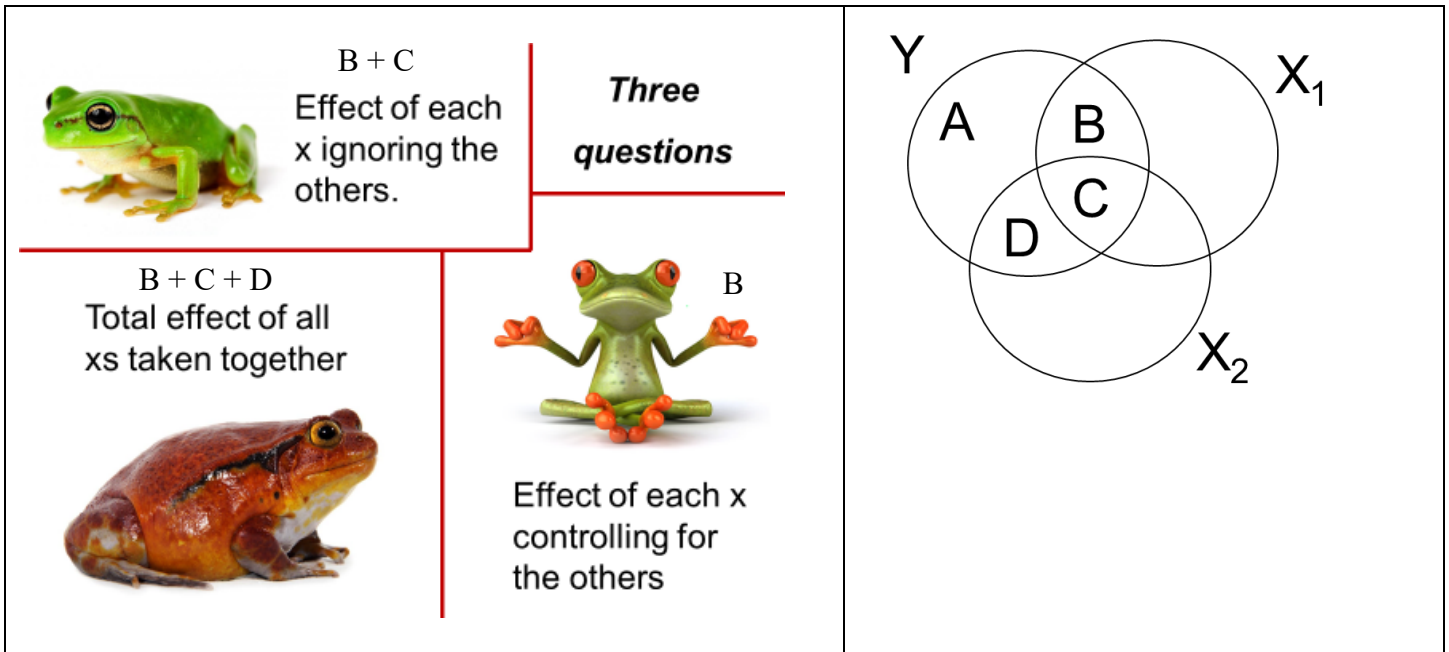
Simple Regression:	Multiple Regression:
Sum of squared error: Measures distance from regression line $\Sigma (y - \hat{y})^2$	Sum of squared error: Measures distance from regression line $\Sigma (y - \hat{y})^2$
Mean square error $\frac{\Sigma(y - \hat{y})^2}{n - 2}$	Mean square error $\frac{\Sigma(y - \hat{y})^2}{n - p - 1}$
Standard Error of Estimate $\sqrt{\frac{\Sigma (y - \hat{y})^2}{n - 2}}$	Standard Error of Estimate $\sqrt{\frac{\Sigma(y - \hat{y})^2}{n - p - 1}}$

p = number of predictors

Partitioning the sum of squares

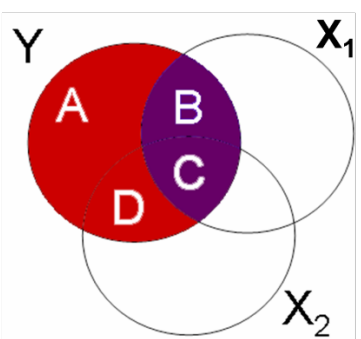
Analysis of Variance (ANOVA)

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained (regression)	Total – Error	p	SS / df	$\frac{MS \text{ Explained}}{MS \text{ Error}}$
Error (residual)		$n - p - 1$	SS / df	
Total		$n - 1$	p = number of predictors	



Effect of each x ignoring the others

Correlation = Standardized effect of each x ignoring the others.

 correlation $r^2 = \frac{B + C}{A + B + C + D}$ $r = \sqrt{\frac{B + C}{A + B + C + D}}$	Significance testing Correlation $t = \frac{r}{sd_r}$ $sd_r = \sqrt{\frac{1 - r^2}{n - 2}}$ Degrees of freedom = $n - 2$
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Effect of all x s taken together

R and R^2

R is the multiple correlation.

R is the correlation between observed y and predicted y .

R^2 indicates the amount of variance in y explained by ALL the x variables combined together.

$$R^2 = \frac{\text{Explained SS}}{\text{Total SS}}$$

$$R = \sqrt{R^2}$$

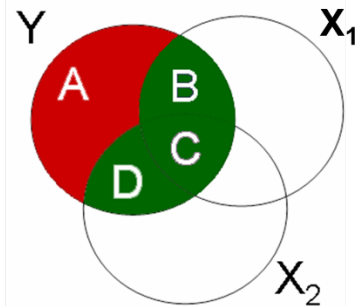
Analysis of Variance (ANOVA)

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained (regression)	Total – Error	p	SS / df	$\frac{MS \text{ Explained}}{MS \text{ Error}}$
Error (residual)		$n - p - 1$	SS / df	
Total		$n - 1$	$p = \text{number of predictors}$	

R , R^2 and r^2 cannot be negative
They are biased (but r is not)

	r	r^2
Study 1	.05	.0025
Study 2	-.1	.01
Study 3	.2	.04
Study 4	-.05	.0025
Study 5	.1	.01
Study 6	-.2	.04
<i>etc.</i>		
Average	0	.0175

Adjusted R^2 is an unbiased estimate of R^2 .



$$R^2 = \frac{B + C + D}{A + B + C + D} \quad R = \sqrt{\frac{B + C + D}{A + B + C + D}}$$

R and R^2

Significance testing

F value from ANOVA table

Effect of each x controlling for the others

Beta

Standardized Beta

Partial Correlation

Semipartial Correlation (R^2 Change)

Beta: The predicted increase in Y for a one-unit increase in X **when all the other variables are held constant.**

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + e$$

If $x_2 = 0$ and $x_3 = 0$

$$y = b_0 + b_1x_1 + b_2(0) + b_3(0) + e$$

$$y = b_0 + b_1x_1 + e$$

Beta: If one or more control variables are extracted out of x and y , beta is the predicted increase in **residual** y for a one-unit increase in **residual** x .

Calculate b_1

$$y = b_0 + b_1 x_1 + b_2 x_2 + e$$

x_2 is control



victims

y x_1

$$y = b_0 + b_2 x_2 + e_{y.x2}$$

Step 1

Extract x_2 out of y .
Create residual score.

Read the period as “controlling for.”

$e_{y.x2}$ = residual of y controlling for x_2

Step 2

$$x_1 = b_0 + b_2 x_2 + e_{x1.x2}$$

Extract x_2 out of x_1 .
Create residual score.

Step 3

Simple regression:

Use residual x to predict residual y .

$$e_{y.x2} = b_0 + b_1 e_{x1.x2} + e$$

Calculate b_2

$$y = b_0 + b_1 x_1 + b_2 x_2 + e$$

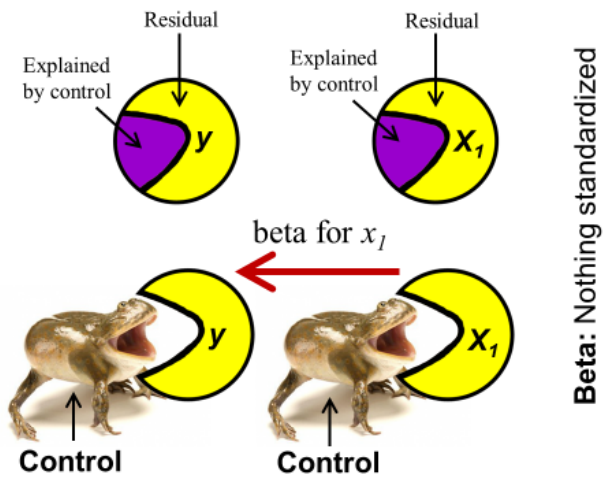
$$y = b_0 + b_1 x_1 + e_{y.x1}$$

$$x_2 = b_0 + b_1 x_1 + e_{x2.x1}$$

$$e_{y.x1} = b_0 + b_2 e_{x2.x1} + e$$

Three ways of addressing the same question:

- **Beta:** no standardization
- **Standardized beta:** Both x and y are standardized prior to analysis.
- **Partial correlation:** The residuals for x and y are standardized.

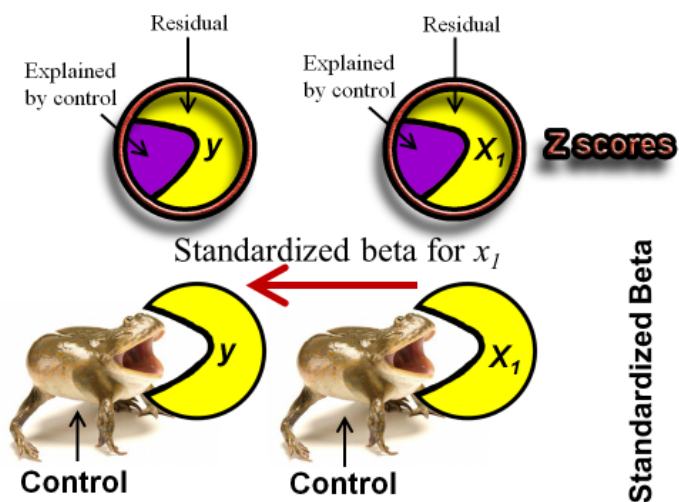


$$\text{Influence} = b_0 + b_1 (\text{income}) + b_2 (\text{smiles}) + e$$

Descriptive Statistics

	Mean	Std. Deviation	N
Influence	40.88	12.74	552
Income	85357.97	57665.06	552
Smiles	2.56	.95	552

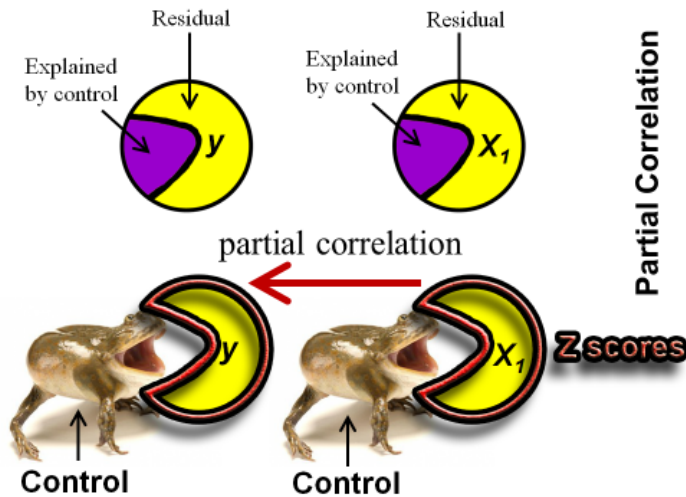
	Beta	Standardized Beta	Partial Correlation
b_0	38.55	0	0
b_1 (income)	.00007	.306	.308
b_2 (smiles)	1.34	.101	.106



Standardized beta: x and y are standardized

- The resulting beta when all variables are standardized prior to analysis.
- Has a scale similar to a correlation.
- Allows you to compare relative UNIQUE contributions of each variable.

$$\text{standardized } b = b \frac{sd_x}{sd_y}$$



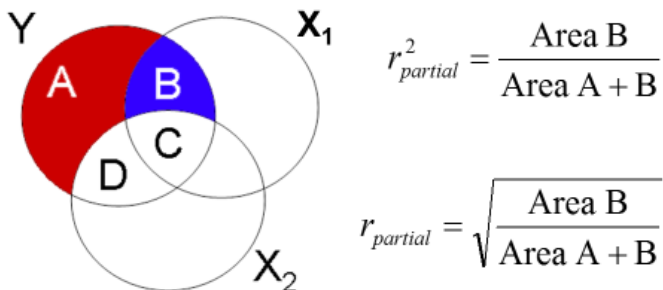
Partial Correlation: residuals of x and y standardized

$$r_{\text{partial}} = \sqrt{\frac{t^2}{t^2 + df}}$$

Use the t value for the beta weight

Use the residual (error) $df = n - p - 1$

If the beta is negative, make the correlation negative



$$r_{\text{partial}} = \sqrt{\frac{\text{Area B}}{\text{Area A} + \text{Area B}}}$$

beta = Area B

Standardized beta

$$= \text{Area B when } sd_Y^2 = sd_{X1}^2 = 1$$

Beta Significance testing

$$t = \frac{b}{sd_{beta}}$$

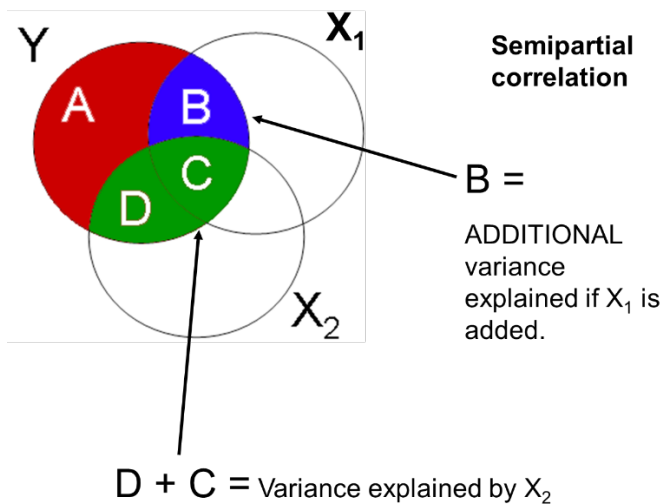
$$df = n - p - 1$$

t -values will be equivalent:

t for beta

t for standardized beta

t for partial correlation

Semipartial correlation, Delta R^2 , ΔR^2 , R^2 change

$$r_{semipartial}^2 = \frac{\text{Area B}}{\text{Area A} + \text{B} + \text{C} + \text{D}}$$

$$r_{semipartial} = \sqrt{\frac{\text{Area B}}{\text{Area A} + \text{B} + \text{C} + \text{D}}}$$

Can be used with sets of predictors

Step 1: Compute Constrained model R squared $y = b_0 + b_2x_2 + e$

Step 2: Compute Full model R squared $y = b_0 + b_1x_1 + b_2x_2 + e$

Step 3: Compute the difference $R^2 \text{ full} - R^2 \text{ constrained}$

DR2 and squared semipartial correlation

- The decrement in R^2 that would result if this variable (or set of variables) was eliminated from the model.
- The increase in R^2 that results from adding this variable (or set of variables) to the model.
- Can be used with multiple steps.

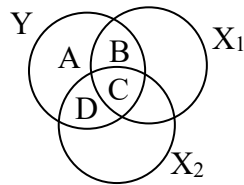
ΔF significance test

	Sum of Squares	Degrees of freedom	Mean square	ΔF
Regression	$F - C$	$F - C$	$\frac{SS_{F-C}}{df_{F-C}}$	$\frac{\text{mean square regression}}{\text{mean square error}}$
Error	Full	Full	Full	

$F - C$ = full minus constrained

Full = insert value from full model

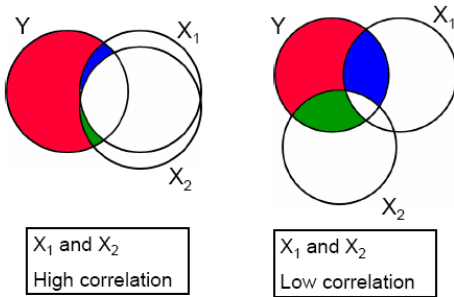
Summary:

3 questions:

Effect of each x ignoring the others	Total effect of all xs taken together	Effect of each x controlling for the others
$B + C$	$B + C + D$	B
$r = \sqrt{\frac{B+C}{A+B+C+D}}$ $r^2 = \frac{B+C}{A+B+C+D}$	$R = \sqrt{\frac{B+C+D}{A+B+C+D}}$ $R^2 = \frac{B+C+D}{A+B+C+D}$	$r_{\text{partial}} = \sqrt{\frac{B}{A+B}}$ $r_{\text{partial}}^2 = \frac{B}{A+B}$ beta = Area B Standardized beta = Area B when $sd_Y^2 = sd_{X1}^2 = 1$ $r_{\text{semipartial}} = \sqrt{\frac{B}{A+B+C+D}}$ $r_{\text{semipartial}}^2 = \frac{B}{A+B+C+D}$

Collinearity (and multi-collinearity)

- Increases standard error of beta decreases significance of beta
- Makes the solution unstable (and maybe even impossible)



Tolerance

- To compute tolerance for X₁
- Regress X₁ on all other X variables and compute the R²
- Tolerance for X₁ is 1-R²

Note that X₁ and X₂ are highly correlated. X₃ is less correlated with the other two X variables. All variables are correlated with Y.

Correlations

		X1	X2	X3	Y
X1	Pearson Correlation	1.000	.995**	.322**	.503**
	Sig. (2-tailed)		.000	.000	.000
	N	124	124	124	124
X2	Pearson Correlation	.995**	1.000	.329**	.513**
	Sig. (2-tailed)	.000		.000	.000
	N	124	124	124	124
X3	Pearson Correlation	.322**	.329**	1.000	.637**
	Sig. (2-tailed)	.000	.000		.000
	N	124	124	124	124
Y	Pearson Correlation	.503**	.513**	.637**	1.000
	Sig. (2-tailed)	.000	.000	.000	
	N	124	124	124	124

** . Correlation is significant at the 0.01 level (2-tailed).

When Both X1 and X2 are entered together, their beta weights have a high error variance. Thus they are untrustworthy and non-significant. They “eat up” each other’s variance.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.519 ^a	.269	.257	.5524

a. Predictors: (Constant), X2, X1

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	13.613	2	6.807	22.307	.000 ^a
	Residual	36.921	121	.305		
	Total	50.535	123			

a. Predictors: (Constant), X2, X1

b. Dependent Variable: Y

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	30.258	3.481		8.692	.000
	X1	-.498	.502	-.800	-.991	.324
	X2	.893	.550	1.309	1.622	.107

a. Dependent Variable: Y

In contrast, note the difference when X1 is entered with X3. Here both variables are significant. Without the problem of collinearity, the model looks much better.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.710 ^a	.505	.497	.4548

a. Predictors: (Constant), X3, X1

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	25.507	2	12.753	61.657	.000 ^a
	Residual	25.028	121	.207		
	Total	50.535	123			

a. Predictors: (Constant), X3, X1

b. Dependent Variable: Y

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	8.598	3.839		2.240	.027
	X1	.207	.042	.333	4.930	.000
	X3	.621	.079	.529	7.835	.000

a. Dependent Variable: Y

If all three variables are entered together, only X3 is significant. This is not because X3 is more important, but rather, because X1 and X2 “eat up” each other’s variance.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.715 ^a	.511	.499	.4537

a. Predictors: (Constant), X3, X1, X2

ANOVA^b

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	25.828	3	8.609	41.817	.000 ^a
	Residual	24.706	120	.206		
	Total	50.535	123			

a. Predictors: (Constant), X3, X1, X2

b. Dependent Variable: Y

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	6.394	4.216		1.516	.132
	X1	-.307	.413	-.493	-.742	.459
	X2	.567	.454	.832	1.250	.214
	X3	.612	.079	.522	7.703	.000

a. Dependent Variable: Y

Suppressor effect

Usually the partial correlation is smaller than the correlation

A suppressor effect occurs when the partial correlation is larger than the correlation

Lecture Add-On Notes

More results Sanford, K. (2005). Attributions and anger in early marriage: Wives are event-dependent and husbands are schematic. <i>Journal of Family Psychology</i>, 19, 180-188. 77 couples 4 conversations	Event dependent <i>Pre-conversation questionnaire</i> Attributions att1 6 item scale Average $\alpha = .80$ Anger hard1 4 item scale Average $\alpha = .90$ Used T1 scores only (not change scores)	Schematic <i>Single assessment</i> Trait Attributions attribs 3 ratings for each of 4 scenarios $\alpha = .87$ Satisfaction das 7 item subscale $\alpha = .79$
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Questions for multiple regression results

1. What is the typical distance scores deviated from the mean of y ?
2. Compute the t statistic for the correlation between x_1 and y ?
3. What is the typical distance observed scores deviated from their predicted values?
4. What is the percent of variance explained?
5. What is the correlation between observed scores and their predicted values?
6. After controlling for the effects of other predictors, how many units is y expected to increase when x_1 increases by 1 unit?
7. After controlling for the effects of other predictors, how many standard deviations is y expected to increase when x_1 increases by 1 standard deviation?
8. Fill in the blanks below with numerical values so that the sentence gives a clear and accurate interpretation of the beta for x_1 .
Y is expected to increase by _____ units if
 x_1 increases by _____ units and
 x_2 is _____ and
 x_3 is _____.
9. Compute the t statistic for each beta.
10. Compute the partial correlation for each predictor.

To answer each of the following questions, provide 5 pieces of information.

11. What is the standardized association between X_1 and Y (without controlling for other variables)?
12. What is the standardized association between X_1 and Y after controlling for other variables in the equation?
13. What is the total percent of variance in Y explained by all the predictors taken together?

Wives

Descriptive Statistics

	Mean	Std. Deviation	N
hard1	4.04	1.26	77
das	4.85	.63	77
attribs	3.11	.83	77
att1	3.08	1.08	77

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.348 ^a	.121	.085	1.204

a. Predictors: (Constant), att1, das, attribs

Correlations

		hard1	das	attribs	att1
hard1	Pearson Correlation	1.000	-.115	.208	.343
	Sig. (1-tailed)	.	.161	.035	.001
	N	77	77	77	77
das	Pearson Correlation	-.115	1.000	-.404	-.373
	Sig. (1-tailed)	.161	.	.000	.000
	N	77	77	77	77
attribs	Pearson Correlation	.208	-.404	1.000	.487
	Sig. (1-tailed)	.035	.000	.	.000
	N	77	77	77	77
att1	Pearson Correlation	.343	-.373	.487	1.000
	Sig. (1-tailed)	.001	.000	.000	.
	N	77	77	77	77

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.28	1.56		1.46	.149
	das	.06	.24	.03	.26	.795
	attribs	.09	.20	.06	.48	.633
	att1	.38	.15	.32	2.52	.014

a. Dependent Variable: hard1

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14.55	3	4.85	3.35	.024 ^a
	Residual	105.75	73	1.45		
	Total	120.30	76			

a. Predictors: (Constant), att1, das, attribs

b. Dependent Variable: hard1

Husbands

Descriptive Statistics

	Mean	Std. Deviation	N
hard1	3.01	1.33	76
das	4.90	.62	76
attribs	2.86	.77	76
att1	2.88	.98	76

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.596 ^a	.356	.329	1.09

a. Predictors: (Constant), att1, attribs, das

Correlations

		hard1	das	attribs	att1
hard1	Pearson Correlation	1.000	-.578	.385	.266
	Sig. (1-tailed)	.	.000	.000	.010
	N	76	76	76	76
das	Pearson Correlation	-.578	1.000	-.456	-.483
	Sig. (1-tailed)	.000	.	.000	.000
	N	76	76	76	76
attribs	Pearson Correlation	.385	-.456	1.000	.415
	Sig. (1-tailed)	.000	.000	.	.000
	N	76	76	76	76
att1	Pearson Correlation	.266	-.483	.415	1.000
	Sig. (1-tailed)	.010	.000	.000	.
	N	76	76	76	76

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	47.16	3	15.72	13.24	.000 ^a
	Residual	85.46	72	1.19		
	Total	132.61	75			

a. Predictors: (Constant), att1, attribs, das

b. Dependent Variable: hard1

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	8.01	1.62		4.95	.000
	das	-1.14	.24	-.53	-4.65	.000
	attribs	.29	.19	.17	1.53	.131
	att1	-.08	.15	-.06	-.54	.591

a. Dependent Variable: hard1

Writing a summary of regression results

In regression analysis, wives' anger during conflict was predicted using their event-dependent attributions, trait attributions, and overall relationship satisfaction. The three predictors explained a significant portion of variance in wives' anger ($R^2 = .12$, $F(3, 73) = 3.35$, $p < .05$). Correlations and standardized betas are reported in Table 1. There was a medium positive correlation between event dependent attributions and anger, and event dependent attributions explained unique variance in anger after controlling for the effects of the other variables in the equation. Although there was a small positive correlation between trait attributions and anger, the effect became non-significant after controlling for other variables. Notably, relationship satisfaction was not significantly related to anger.

Table 1.

Predicting wives' anger during marital conflict

	Correlation	Standardized beta
Satisfaction	-.12	.03
Trait attributions	.21*	.06
Event dependent attributions	.34***	.32*

* $p < .05$, ** $p < .01$, *** $p < .001$

Notes:

The homework for this unit provides details on how written summaries are scored.

In written summaries, numerical values are usually easiest for readers to interpret if they are rounded to two decimals. However, if $p < .01$, then round the p -value to three decimals, and if $p < .001$ then print " $p < .001$." NEVER print " $p = .000$ " because it is not possible for a p to be exactly zero and it looks odd if you round the p to an impossible value.

The table is an important part of your summary. The video provides instructions to draw three horizontal lines on the table. This is important when you create a table in a word document for a project, but you will NOT be expected to draw horizontal lines when answering learning assessment questions on Canvas.

Data taken from: Sanford, K. (In Press). The Couples Emotion Rating Form: Psychometric properties and theoretical associations. <i>Psychological Assessment</i>. Hard emotion and Soft emotion Website questionnaire • 377 married women • 266 married men Results from participants reporting on a conflict that occurred within the last two weeks.	Outcome variable Positive communication 7 items <i>Examples:</i> I politely talked about my feelings. I carefully listened so I could understand my partner. Alpha = .85 Communication Patterns Questionnaire (CPQ) Relationship conflict 7 items Alpha = .78 Quality Marital Index (QMI) Relationship Satisfaction 7 items Alpha = .78	Partner soft emotion Sad hurt concerned disappointed Alpha = .78 Partner hard emotion Angry Annoyed Irritated aggravated Alpha = .83 Partner flat emotion bored uninterested indifferent disengaged Alpha = .88
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Hard emotion

- Increased negative communication
- Decreased positive communication
- Likely to be perceived as threatening
- Self-focused goals

Soft emotion

- Increased positive communication
- May slightly *increase* negative communication
- Likely to be perceived as showing vulnerability
- Relationship focused goals

Flat emotion

- Withdrawal from conflict
- May not lead to negative or positive communication.
- Long term effects may be detrimental.

Descriptive Statistics

	Mean	Std. Deviation	N
pos	1.83	.76	377
CPQ	3.39	1.73	377
qmi	4.75	1.32	377
p_hard	2.57	1.02	377
p_soft	1.54	.97	377
p_flat	.96	.95	377

Wives page 1

Correlations

		pos	CPQ	qmi	p_hard	p_soft	p_flat
pos	Pearson Correlation	1.00	-.28	.23	-.19	.23	-.08
	Sig. (1-tailed)	.	.000	.000	.000	.000	.057
	N	377	377	377	377	377	377
CPQ	Pearson Correlation	-.28	1.00	-.67	.40	-.05	.24
	Sig. (1-tailed)	.000	.	.000	.000	.182	.000
	N	377	377	377	377	377	377
qmi	Pearson Correlation	.23	-.67	1.00	-.37	.10	-.25
	Sig. (1-tailed)	.000	.000	.	.000	.024	.000
	N	377	377	377	377	377	377
p_hard	Pearson Correlation	-.19	.40	-.37	1.00	-.05	-.02
	Sig. (1-tailed)	.000	.000	.000	.	.158	.329
	N	377	377	377	377	377	377
p_soft	Pearson Correlation	.23	-.05	.10	-.05	1.00	-.25
	Sig. (1-tailed)	.000	.182	.024	.158	.	.000
	N	377	377	377	377	377	377
p_flat	Pearson Correlation	-.08	.24	-.25	-.02	-.25	1.00
	Sig. (1-tailed)	.057	.000	.000	.329	.000	.
	N	377	377	377	377	377	377

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.29 ^a	.08	.08	.73	.08	17.22	2	374	.000
2	.37 ^b	.14	.12	.71	.05	7.28	3	371	.000

a. Predictors: (Constant), qmi, CPQ

b. Predictors: (Constant), qmi, CPQ, p_soft, p_flat, p_hard

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	18.43	2	9.21	17.22	.000 ^a
	Residual	200.10	374	.54		
	Total	218.53	376			
2	Regression	29.55	5	5.91	11.60	.000 ^b
	Residual	188.97	371	.51		
	Total	218.53	376			

a. Predictors: (Constant), qmi, CPQ

b. Predictors: (Constant), qmi, CPQ, p_soft, p_flat, p_hard

c. Dependent Variable: pos

Wives page 2**Coefficients^a**

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.96	.26		7.43	.000
	CPQ	-.10	.03	-.23	-3.47	.001
	qmi	.05	.04	.08	1.17	.242
2	(Constant)	1.88	.29		6.45	.000
	CPQ	-.10	.03	-.23	-3.33	.001
	qmi	.03	.04	.04	.66	.511
	p_hard	-.05	.04	-.07	-1.27	.206
	p_soft	.17	.04	.22	4.41	.000
	p_flat	.03	.04	.04	.70	.484

a. Dependent Variable: pos

Descriptive Statistics

	Mean	Std. Deviation	N
pos	1.92	.75	266
CPQ	3.34	1.51	266
qmi	4.87	1.23	266
p_hard	2.47	.98	266
p_soft	1.89	.93	266
p_flat	.68	.79	266

Husbands page 1

Correlations

		pos	CPQ	qmi	p_hard	p_soft	p_flat
pos	Pearson Correlation	1.00	-.30	.18	-.14	.30	-.12
	Sig. (1-tailed)	.	.000	.002	.009	.000	.021
	N	266	266	266	266	266	266
CPQ	Pearson Correlation	-.30	1.00	-.67	.31	.02	.33
	Sig. (1-tailed)	.000	.	.000	.000	.373	.000
	N	266	266	266	266	266	266
qmi	Pearson Correlation	.18	-.67	1.00	-.23	-.02	-.36
	Sig. (1-tailed)	.002	.000	.	.000	.387	.000
	N	266	266	266	266	266	266
p_hard	Pearson Correlation	-.14	.31	-.23	1.00	.03	-.04
	Sig. (1-tailed)	.009	.000	.000	.	.320	.267
	N	266	266	266	266	266	266
p_soft	Pearson Correlation	.30	.02	-.02	.03	1.00	-.05
	Sig. (1-tailed)	.000	.373	.387	.320	.	.209
	N	266	266	266	266	266	266
p_flat	Pearson Correlation	-.12	.33	-.36	-.04	-.05	1.00
	Sig. (1-tailed)	.021	.000	.000	.267	.209	.
	N	266	266	266	266	266	266

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.31 ^a	.09	.09	.72	.09	13.57	2	263	.000
2	.44 ^b	.19	.17	.68	.10	10.36	3	260	.000

a. Predictors: (Constant), qmi, CPQ

b. Predictors: (Constant), qmi, CPQ, p_soft, p_hard, p_flat

ANOVA^a

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	13.93	2	6.96	13.57	.000 ^a
	Residual	134.96	263	.51		
	Total	148.88	265			
2	Regression	28.33	5	5.67	12.22	.000 ^b
	Residual	120.55	260	.46		
	Total	148.88	265			

Husbands page 2

a. Predictors: (Constant), qmi, CPQ

b. Predictors: (Constant), qmi, CPQ, p_soft, p_hard, p_flat

c. Dependent Variable: pos

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.61	.34		7.67	.000
	CPQ	-.17	.04	-.34	-4.23	.000
	qmi	-.03	.05	-.05	-.60	.552
2	(Constant)	2.28	.36		6.39	.000
	CPQ	-.16	.04	-.32	-4.03	.000
	qmi	-.03	.05	-.05	-.70	.485
	p_hard	-.05	.05	-.07	-1.13	.259
	p_soft	.25	.05	.30	5.44	.000
	p_flat	-.03	.06	-.03	-.44	.658

a. Dependent Variable: pos

Questions for multiple regression results involving two steps

1. Calculate the sum of squares for the explained variance for ΔF .
2. Calculate the degrees of freedom for the explained variance for ΔF .
3. Calculate the mean square for the explained variance for ΔF .
4. What is the mean square error variance for ΔF ?
5. Calculate the ΔF .

To answer each of the following questions, provide 5 pieces of information.

6. What is the standardized association between X3 and Y (without controlling for other variables)?
7. What is the standardized association between X3 and Y after controlling for other variables in the equation?
8. What percent of the variance in Y was explained when just the first set of variables were entered on step one?
9. How much additional variance in Y was explained when the second set of variables was added to the equation on step two?
10. What is the total percent of variance in Y explained by all the predictors taken together?

The following section is not needed for this class.

The final section of the lecture add-on video includes instructions on how to write a summary of regression results **with multiple steps**, but this section is not included as part of our class. For this class, you will **ONLY** need to write a summary of regression results (described earlier), but you do **NOT** need to write a summary of regression results **with multiple steps** (which is depicted below just for reference).

Writing a summary of regression results with multiple steps

In the first step of a regression analysis, wives' positive communication was predicted using their reported level of relationship conflict and their overall relationship satisfaction. The two predictors explained a significant portion of variance in positive communication ($R^2 = .08$, $F(2, 374) = 17.22$, $p < .001$). There was a small negative correlation between conflict and positive communication, and conflict explained unique variance in positive communication after controlling for satisfaction. Although there was a small positive correlation between satisfaction and positive communication, the effect became non-significant after controlling for conflict.

In the second step, ratings of partner soft emotion, partner hard emotion, and partner flat emotion were added to the equation. These three variables explained a significant portion of additional variance in positive communication ($\Delta R^2 = .05$, $\Delta F(3, 371) = 7.28$, $p < .001$), and all five variables taken together explained 14 percent of the variance. Correlations and standardized betas are reported in Table 3. There was a small positive correlation between partner soft emotion and positive communication, and soft emotion explained unique variance in communication after controlling for the effects of the other variables in the equation. Although there was a small negative correlation between partner hard emotion and positive communication, this effect became non-significant after controlling for other variables. Notably, partner flat emotion was not significantly related to positive communication.

Table 3.

Predicting wives' positive communication

	Correlation	Standardized beta
Relationship Conflict	-.28***	-.23***
Relationship Satisfaction	.23***	.04
Partner hard emotion	-.19***	-.07
Partner soft emotion	.23***	.22***
Partner flat emotion	-.08	.04

* $p < .05$, ** $p < .01$, *** $p < .001$